

Quiz 03 - Runtime Function

Instructions:

- Your solution must be submitted in the Quiz03 directory of your Quizzes directory in your GitHub repository.
- Your submission must be a txt file.
- Cheating of any kind is prohibited and will not be tolerated.
- Violating and/or failing to follow any of the rules will result in an automatic zero (0) for the lab.

Grading

Question	Maximum Points	Points Earned
1	0.5	
Total	0.5	

Question 1

For the function below, calculate its runtime function by first stating its input size and creating its runtime table. Make the cost of each statement equal to 1.

```
double P(vector<double>& v)
{
    if(v.length() == 1)
    {
        return 0;
    }
    double mn = v[0];
    double mx = v[1];

    if(mn > mx)
    {
        mn = mn + mx;
        mx = mn - mx;
        mn = mn - mx;
    }

    for(int i = 2; i < v.length(); i += 1)
    {
        if(mx < v[i])
        {
            mx = v[i];
        }
        else if(mn > v[i])
        {
            mn = v[i];
        }
    }
    return (mx - mn);
}
```